

PORT COLBORNE, ON, Canada

WMO: 714630

Lat: 42.867N

Lon: 79.251W

Elev: 602

StdP: 14.38

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.9	(c) 7.6	(d) -5.9	(e) 4.1	(f) 5.2	(g) -1.4	(h) 5.2	(i) 9.9	(j) 48.6	(k) 27.3	(l) 44.8	(m) 27.4	(n) 14.1	(o) 270	(p) 0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.9	79.0	70.0	77.3	70.1	76.0	69.3	74.2	76.4	72.6	75.4	71.3	74.5	10.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.5	127.5	75.7	71.7	119.6	74.5	70.1	113.2	73.7	38.2	76.6	36.7	75.6	35.5	74.6	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 39.9	(b) 34.4	(c) 29.3	(e) -0.9	(f) 83.9	(g) 6.3	(h) 2.9	(i) -5.5	(j) 86.0	(k) -9.2	(l) 87.7	(m) -12.7	(n) 89.3	(o) -17.3	(p) 91.4		
(4) 39.9	(5) 34.4	(6) 29.3	(7) -0.9	(8) 83.9	(9) 6.3	(10) 2.9	(11) -5.5	(12) 86.0	(13) -9.2	(14) 87.7	(15) -12.7	(16) 89.3	(17) -17.3	(18) 91.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.1	26.1	26.3	31.8	42.5	53.4	64.0	69.9	70.1	64.2	53.2	41.9	32.2
	DBStd	17.58	10.02	9.59	8.05	6.94	6.25	5.32	4.06	4.14	6.19	7.67	8.05	8.46
	HDD50	3115	741	664	565	240	35	0	0	0	1	55	262	552
	HDD65	6622	1206	1084	1030	674	365	79	8	8	87	371	694	1016
	CDD50	2416	0	0	0	15	139	421	618	623	426	155	18	1
	CDD65	450	0	0	0	0	5	50	161	166	63	5	0	0
	CDH74	865	0	0	0	2	12	71	280	381	118	1	0	0
	CDH80	37	0	0	0	0	1	5	14	11	6	0	0	0
Wind	WSAvg	13.5	17.2	15.0	13.0	12.4	11.7	10.6	11.2	11.3	12.1	14.7	16.2	16.6
Precipitation	PrecAvg	38.1	3.1	2.1	2.7	3.3	3.3	2.6	3.3	3.1	3.8	3.5	3.7	3.0
	PrecMax	45.9	5.3	4.4	5.0	5.6	6.1	6.7	7.3	5.6	7.5	7.8	5.5	6.1
	PrecMin	31.3	1.6	1.1	1.4	1.6	0.8	0.7	1.1	1.2	1.5	1.4	1.7	1.2
	PrecStd	4.7	1.1	0.7	1.0	1.1	1.5	1.5	1.7	1.2	1.5	1.6	0.9	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.9	48.4	55.6	66.9	75.2	79.6	81.7	81.6	80.3	72.3	61.3	53.4
		MCWB	48.1	44.1	47.6	50.4	60.6	66.9	67.7	68.5	70.7	64.6	55.6	49.9
	2%	DB	45.5	44.3	49.4	60.6	69.6	75.9	78.3	79.0	76.8	69.5	58.4	49.1
		MCWB	43.0	40.5	43.9	48.6	58.7	66.3	69.4	71.3	70.5	63.6	54.0	45.8
	5%	DB	42.0	41.0	45.4	55.6	65.7	73.2	76.6	77.4	75.0	67.0	55.8	46.4
		MCWB	39.3	37.7	40.6	46.6	57.3	65.9	69.8	70.9	69.0	61.1	50.8	43.6
	10%	DB	39.1	37.7	42.1	51.7	62.2	71.1	75.3	76.1	73.1	64.1	53.4	43.2
		MCWB	36.5	34.9	38.1	44.8	55.6	65.3	69.1	69.9	67.3	58.9	48.6	40.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.8	45.5	49.2	55.5	65.2	72.7	75.6	77.0	74.0	67.4	57.5	50.8
		MCDB	50.2	47.3	53.3	62.3	71.8	75.1	76.8	77.9	76.5	70.3	59.9	53.8
	2%	WB	43.2	41.2	44.9	51.4	62.0	70.2	73.4	75.1	72.3	64.7	54.9	46.2
		MCDB	45.1	43.4	48.2	57.9	66.9	72.8	75.8	76.9	75.7	68.3	57.7	48.4
	5%	WB	39.8	37.8	41.6	48.4	59.3	68.6	72.1	73.1	70.3	62.2	51.5	43.6
		MCDB	41.7	40.3	45.0	53.4	64.5	71.8	74.8	75.8	74.0	65.8	55.1	46.1
	10%	WB	36.7	35.0	38.4	45.8	56.5	66.8	70.9	71.5	68.3	59.5	48.9	40.2
		MCDB	38.9	37.4	41.5	50.7	61.0	70.2	74.0	74.8	72.2	63.7	53.0	42.7
Mean Daily Temperature Range	5% DB	MDBR	10.6	10.9	10.7	12.1	11.0	9.0	8.2	8.9	10.1	9.8	9.6	8.8
		MCDBR	12.3	13.2	15.0	19.2	15.9	11.8	9.3	9.4	10.3	11.2	11.6	11.0
		MCWBR	11.7	11.3	10.9	11.7	8.9	7.1	6.9	7.1	7.0	8.3	10.4	10.6
	5% WB	MCDBR	12.5	12.9	14.3	17.0	13.8	9.8	7.4	7.7	9.3	10.4	10.9	11.1
		MCWBR	12.4	11.7	11.6	12.3	9.6	7.5	6.6	7.5	7.4	9.2	10.6	10.8
Clear-Sky Solar Irradiance	taub	0.301	0.335	0.335	0.376	0.405	0.434	0.437	0.425	0.382	0.362	0.315	0.301	
	taud	2.401	2.288	2.391	2.324	2.290	2.243	2.253	2.277	2.376	2.408	2.508	2.457	
	Ebn at Noon	265	271	288	282	276	267	265	264	268	259	260	255	
	Edh at Noon	25	33	33	39	41	43	43	40	34	29	22	22	
All-Sky Solar Radiation	RadAvg	419	671	1034	1399	1716	1862	1874	1661	1318	799	495	330	
	RadStd	38	63	120	149	178	139	143	90	121	77	71	42	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page